ModelIt!

Teacher's Guide

Biomimicry Design Lab

Engineering Solutions Inspired by Nature

Quick Reference

Grade Level	Grade NE
NGSS Standard	MS-LS4-4, MS-ETS1-1, MS-ETS1-2, MS-ETS1-3, MS-ETS1-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS4-4, MS-ETS1-1, MS-ETS1-2, MS-ETS1-3, MS-ETS1-4

Construct an explanation based on evidence that describes how genetic variations of traits in a population increase some individuals' probability of surviving and reproducing. Define the criteria and constraints of a design problem. Evaluate competing design solutions. Analyze data from tests to determine similarities and differences. Develop a model to generate data for iterative testing.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a 7-component model to evaluate competing biomimicry designs across multiple performance metrics.
Disciplinary Core Idea	**LS4.C Adaptation** — Animal features are adaptations shaped by natural selection — understanding their function is essential to effective biomimicry.
Crosscutting Concept	**Structure and Function** — The structure of an animal feature determines its function — and replicating that structure in an engineering context requires understanding WHY the shape works, not just WHAT it looks like.

CAST Testing Alignment

MULTIPLE CHOICE:

An engineer designs two underwater robots: Robot A uses silicone skin (lifelike) and Robot B uses hard plastic. Both have the same power source. According to the tradeoff model, which statement is MOST LIKELY true? A) Robot A has better durability B) Robot B has lower detection risk C) Robot A uses more energy D) Both have identical performance

CONSTRUCTED RESPONSE:

A wildlife research team needs a robot bird to observe eagle nests. Using your model, recommend a design approach. Which animal feature should they copy first? What material and power tradeoffs should they consider? Use evidence from your model simulations.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

This lesson builds on Level 1's simple tail-shape model by expanding to a full engineering design system with seven interacting components. Students now experience the full complexity of biomimicry design decisions.

The Science of Biomimicry

Biomimicry has produced remarkable innovations: Velcro (burrs), bullet trains (kingfisher beaks), solar panels (leaf photosynthesis), water collection (Namib beetle), adhesives (gecko feet), and more. In each case, engineers studied HOW a natural feature works, then adapted the PRINCIPLE (not an exact copy) for human use.

Real Animal Robots

Engineers have built robotic fish (MIT's SoFi, which swims alongside real fish), robotic birds (Festo's SmartBird), robotic insects (Harvard's RoboBee), and robotic snakes (Carnegie Mellon). These bio-inspired robots are used for environmental monitoring, search and rescue, and scientific research.

The Tradeoff Challenge

Every engineering design involves tradeoffs. Making a robot more lifelike requires softer, more flexible materials — but these are less durable. Larger batteries provide more runtime — but add weight that reduces mobility. Faster movement looks more natural — but drains power quickly. There is no perfect design, only optimal balances for specific use cases.

Evolution as R&D

Evolution through natural selection has been 'testing designs' for 3.8 billion years. Every living organism represents a solution that has been refined over millions of generations. This is why biomimicry works — nature has already explored the design space more thoroughly than any human R&D department ever could.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Animal Feature, Material Choice, and Power Source as external design decisions. Movement Accuracy, Energy Use, Durability, and Detection Risk are internal performance outcomes.

E **Establish Relationships**
Students discover complex relationships: better materials improve accuracy and reduce detection risk but increase energy use and decrease durability. More power improves all performance metrics but has diminishing returns and adds weight.

V **Visualize & Model**
Students build the 7-component model with three inputs feeding four outputs through interconnected relationships.

E **Evaluate Outcomes**
Students run four scenarios testing different design philosophies and comparing outcomes across all four metrics.

R **Revise & Extend**
Students add Cost, Environmental Resistance, or Noise Level to see how real-world constraints further complicate design decisions.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Remember Level 1 — we learned that flat tails swim best. NOW we design a complete animal robot. Can we fool real animals?"*

Watch For

Show video of MIT's SoFi robot fish swimming with real fish in coral reefs.

Slide 2: Learning Objectives

3 min

Say: *"Today we tackle the FULL design challenge — not just one feature, but the whole robot."*

Watch For

Pre-teach 'fidelity,' 'constraint,' and 'iteration.' Connect to Level 1 vocabulary.

Slide 3: Big Question

3 min

Say: *"What would it take to build a robot so lifelike that a real beaver thinks it is another beaver?"*

Watch For

Brainstorm in groups: List everything a robot would need to fool a real animal (appearance, movement, sound, smell?).

Slide 4: LEVER Framework

2 min

Say: *"This model has TRADEOFFS — improving one thing makes another worse. Welcome to real engineering."*

Watch For

Preview the concept of tradeoffs with a simple example: a heavier backpack carries more but is harder to walk with.

Slide 5: Activity 1: Components

8 min

Say: *"Three design decisions you make, four performance outcomes you measure. Sort them."*

Watch For

Guide sorting. Emphasize: the externals are CHOICES, the internals are CONSEQUENCES of those choices.

Slide 6: Activity 2: Connections

10 min

Say: *"This is where it gets interesting. Better materials improve accuracy BUT also increase energy use. That is a tradeoff."*

Watch For

Draw relationships. Key: some arrows are positive AND negative from the same input. Highlight the tradeoff tension.

Slide 7: Activity 3: Simulation

12 min

Say: *"Four designs, four philosophies. High fidelity, budget build, power boost, and wrong feature. Which wins?"*

Watch For

Run all four scenarios. Create a comparison table. Key moment: wrong feature scenario shows that fundamentals beat materials.

Slide 8: Discoveries

4 min

Say: *"There is NO perfect design. But there are SMART designs that balance tradeoffs for the specific mission."*

Watch For

Summarize the key tradeoffs. Discuss: Which design would YOU choose for a specific mission?

Slide 9: STEM Challenge

12 min

Say: *"Now design YOUR animal robot. Every choice must be justified with model data."*

Watch For

Distribute sketchbook materials. Students choose an animal, research adaptations, and design a robot with tradeoff analysis.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Animal Feature Selected, Material Choice, Power Source
- **INTERNAL:** Movement Accuracy, Energy Use, Durability, Detection Risk

Relationships

- **Animal Feature to Movement Accuracy:** VARIES — Choosing the right feature (e.g., flat tail for swimming) dramatically improves movement accuracy. The wrong feature (pointed tail in water) cannot be compensated for by other choices.
- **Material Choice to Movement Accuracy:** POSITIVE (+) with more lifelike materials — More flexible, lifelike materials (silicone, soft polymers) allow more natural movement. Rigid materials (hard plastic, metal) create stiff, unnatural motion.
- **Material Choice to Detection Risk:** NEGATIVE (-) with more lifelike materials — More lifelike materials reduce detection risk — they look and feel more like real animal tissue. Hard, shiny materials are immediately identifiable as artificial.
- **Material Choice to Energy Use:** POSITIVE (+) with more lifelike materials — Softer, more lifelike materials require more energy to actuate — they absorb energy that rigid materials would transmit directly. Moving a soft silicone tail takes more power than a hard plastic one.
- **Material Choice to Durability:** NEGATIVE (-) with more lifelike materials — Soft, flexible materials wear out faster than rigid ones. Silicone tears, soft polymers degrade in UV light, while hard plastic lasts years.
- **Power Source to all outputs:** GENERALLY POSITIVE — More power improves movement accuracy and reduces detection risk (smoother movement), but has diminishing returns and adds weight that can increase detection risk if the robot becomes unnaturally heavy.

Key Discoveries

- The fundamental design choice (which animal feature to copy) matters MORE than materials or power
- Lifelike materials reduce detection risk but increase energy use and decrease durability
- More power helps but creates its own problems — heavier batteries, more heat, shorter lifespan
- No single design maximizes ALL outputs — engineering is about finding the BEST BALANCE of tradeoffs
- Nature's designs are efficient because evolution has been optimizing for millions of years

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The most expensive design is always the best"	Expensive materials and large power sources do not guarantee the best performance. The 'wrong feature' scenario proves that a fundamentally flawed design cannot be saved by premium components. The best design is the one that makes the RIGHT fundamental choices, then uses appropriate (not necessarily expensive) materials to execute them.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Robots need to look exactly like animals to fool them"	Most animals rely primarily on MOVEMENT and BEHAVIOR, not visual appearance. A robot that moves like a real animal but looks slightly different may fool animals better than a photorealistic robot that moves unnaturally. Movement accuracy is often more important than visual fidelity.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"More power always makes robots better"	The power boost scenario shows diminishing returns. Doubling battery size does NOT double performance. Extra power adds weight, heat, and size — all of which can INCREASE detection risk. Real engineers optimize for efficiency (maximum output per unit of energy), not raw power.